

On The Quantitative Importance of Non-fundamentalness for News Shocks

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October 2014

Abstract

It has been rightly suggested (see e.g. Forni and Gambetti [2014] and Forni, Gambetti, and Sala [2014]) that the identification of news shocks can be subject to non-fundamentalness. In this note we first illustrate why non-fundamentalness is not an either/or question, but is a quantitative issue which can be more or less severe. Second, we use the approach suggested by Forni, Gambetti, and Sala [2014] to re-examine whether non-fundamentalness is likely an important problem for the identification of news shocks. Using our preferred strategy for identifying news shocks, we find that non-fundamentalness is quantitatively unimportant and that news shocks continue to generate significant business cycle type fluctuations when adjust the estimating procedure to take into account the potential non-fundamentalness issue.

Key Words : News, Business Cycles, Non-Fundamentalness

JEL Class. : E3

1 Introduction

In a very interesting work, Forni, Gambetti, and Sala [2014] have addressed the important question of whether the identification of news shocks is subject to a non-fundamentalness problem.¹ Non-fundamentalness arises when the economic variables used by an econometrician do not contain enough information to recover the desired shocks. In such a case, the

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‡We thank Mario Forni, Luca Gambetti and Luca Sala for sharing their codes and data and for their comments. We also thank Patrick Fève for comments.

¹The first critique of structural VAR models on the basis of potential non-fundamentalness is Lippi and Reichlin [1993]. See also Lütkepohl [2012] in the case of the identification of news shocks.

economic variables do not have a VAR representation in the structural shocks, implying that such shocks cannot be obtained by estimating a VAR with these variables.

In order to explore the importance of non-fundamentalness for the identification of news shocks, Forni, Gambetti, and Sala [2014] use a dataset composed of 107 US quarterly macroeconomic series and estimate its principal components. They begin by identifying news shocks following the identification strategy proposed in Barsky and Sims [2011] and show that the resulting estimated shocks are not orthogonal to the estimated principal components, which is evidence for non-fundamentalness. They then explain why estimating an augmented VAR that includes principal components from a large data set offers a means of overcoming a non-fundamentalness problem. Using this procedure combined with the identification strategy of Barsky and Sims [2011], Forni, Gambetti, and Sala [2014] find that the resulting impulse response for news shocks do not generate business type fluctuations and accordingly conclude that there are no new in business cycles.

Non-fundamentalness is clearly an important issue in time series econometrics. However, one needs to distinguish between the qualitative importance of this issue from its quantitative importance. As we have shown in Beaudry and Portier [2014], on a qualitative ground, a model with news shocks may give rise to a non-fundamental representation but does not necessarily do so. It depends on the precise properties of the news process as well as the variables available to the econometrician. But more importantly for the discussion at hand, even when a model with news shocks gives rise to a non-fundamental representation, its fundamental representation can be very “close” to its non-fundamental one, implying that VAR methods may nonetheless deliver a good approximation of structural impulse responses even when the non-fundamentalness problem arises.

In this note, we make two contributions. First, we argue that the R^2 associated with the test for non-fundamentalness proposed by Forni, Gambetti, and Sala [2014] may be a better indication of the relevance of the non-fundamentalness problem than the test itself. Second, we illustrate this claim by considering the identification of news shocks.

Specifically, we begin by illustrating why non-fundamentalness should be viewed as a quantitative issue not an either/or issue. To do so, we use a simple Lucas’ tree model with

news. We show that for reasonable discount factors non-fundamentality is not a quantitative problem in this setup. Nonetheless, we show that the test for non-fundamentality proposed in Forni, Gambetti, and Sala [2014] will detect significant non-fundamentality even when its economic importance is minor. We argue that the R^2 associated with the test may be a better indication of the relevance of the non-fundamentality problem than the test itself. We are not the first to recognize that standard SVAR methods can work well in identifying shocks even when these shocks are non-fundamental. This quantitative point has been most clearly in Sims [2012] which looks at the issue in a quantitatively reasonable DSGE model.²

We then turn to the issue of whether or not the identification of technological news shock is subject to a quantitatively important non-fundamentality problem. We replicate the procedure suggested by Forni, Gambetti, and Sala [2014] of including the principal components extracted from their large data set into an SVAR as a means of overcoming the potential non-fundamentality problem. However we use a different identification scheme; one that we have argued elsewhere is a robust way of identifying technological news shocks (see Beaudry and Portier [2014] and Portier [2014]). In our procedure, which extends the logic Beaudry and Portier [2006], a news shock is identified as the only shock that can have a permanent effect on TFP but that does not have an impact effect on TFP.³ With this identification strategy, and using the factors of Forni, Gambetti, and Sala [2014], we first show that the identified news shock does not pass the fundamentality test they proposed. However, we find that the R^2 of this regression is small, which our theoretical simulations would lead us to believe that the non-fundamentality problem may be minor. Accordingly, when we re-estimate our VARs with the principal components included, we find that the impulse re-

²Beaudry, Portier, and Seymen [2013] have also shown that the two prominent structural VAR approaches in the news literature (Beaudry and Portier [2006] and Barsky and Sims [2011]) are in general capable of recuperating news shock dynamics from artificially generated data.

³The object of this note is not to debate whether our preferred strategy for the identification of news shocks is better or not to that proposed in Barsky and Sims [2011]. The issue is whether previous results which have suggested the importance of news shocks in business cycles could have been misleading due to the problem of non-fundamentality. For this argument to be relevant it should be applied to identification strategies that have provided robust evidence in support of news shocks (as the one we use here), not to an identification strategy such as used by Barsky and Sims [2011] that have suggested that such shocks may be rather unimportant even when disregarding the non-fundamentality problem. See Beaudry, Nam, and Wang [2011] for an in depth discussion of the robustness of Barsky and Sims's [2011] results

sponses associated with news shocks are not significantly modified. In fact, the news shock obtained after including the principal components to the VAR continue to exhibit business cycle type fluctuations, as consumption, hours and investment increase in the short run, while TFP increases only in the long run. We therefore conclude that non-fundamentalness may not be important for the identification of news shocks and that there remains news in business cycles.

2 On the severity of the non-fundamentalness problem in a stylized model

In this section, we simulate the simple Lucas' tree model with news presented in Forni, Gambetti, and Sala [2014]. In this model, non-fundamentalness is always present. We use this model to show that for modest to high values of the discount factor, the non-fundamentalness is not quantitatively important.

2.1 A simple Lucas' tree model with news

Consider an endowment economy with one representative tree and one representative household. The tree produces each period a dividend Θ_t , that we call TFP. The process of Θ has a news component ε :

$$\Theta_t = \Theta_{t-1} + \varepsilon_{t-2} + \eta_t,$$

where ε and η are gaussian with unit variance ⁴. Preferences are given by

$$E_0 \sum_{t=0}^{\infty} \beta^t c_t,$$

such that the equilibrium price of a tree, that we call SP for stock prices, is given, after solving the model, by

$$SP_t = \frac{\beta}{1-\beta} \Theta_t + \frac{\beta}{1-\beta} (\beta \varepsilon_t + \varepsilon_{t-1})$$

The structural moving average (MA) representation of the solution is given by

$$\begin{pmatrix} \Delta \Theta_t \\ \Delta SP_t \end{pmatrix} = \begin{pmatrix} L^2 & 1 \\ \frac{\beta^2}{1-\beta} + \beta L & \frac{\beta}{1-\beta} \end{pmatrix} \begin{pmatrix} \varepsilon_t \\ \eta_t \end{pmatrix}.$$

⁴Our results are robust to a change of the TFP process (longer or shorter news) and to the relative size of the variances.

The determinant of the MA coefficients matrix is

$$-\frac{\beta^2}{1-\beta} - \beta L + \frac{\beta}{1-\beta} L^2.$$

The roots of that determinant are 1 and $-\beta$. We assume that only current and past values of TFP and SP are observed by the econometrician. As $|\beta| < 1$, the shocks ε_t and η_t are non-fundamental for the variables $\Delta\Theta_t$ and ΔSP_t , and cannot be recovered by an econometrician. Note that when β goes to one, in economic terms when the stock price weights a lot future Θ , so that we get closer to the fundamental case. What do we mean by “closer”? The metric we will consider is the distance between the estimated impulse response functions to the identified news shock $\tilde{\varepsilon}$ and their theoretical counterpart.

2.2 Simulation results

We simulate the above model, and recover the observables TFP and SP. We then estimate a bivariate VAR and use the simple Beaudry and Portier [2006] short run identification scheme. The news is identified as the shock that has no effect on impact on TFP. We then compare the IRF of TFP and SP to a news shock, and perform the Forni, Gambetti, and Sala [2014] test for non-fundamentalness. The test consists in projecting $\tilde{\varepsilon}$ (the identified news shock) on a constant and the past of the true ε and η that have been used to generate the data. As the shocks are *i.i.d.* and orthogonal one to each other, the R^2 of that regression should be 0 if $\tilde{\varepsilon}$ was indeed equal to ε .

We consider three values for β : .01, .7 and .99. The non-fundamentalness problem is expected to be “less severe” when β is large. We consider estimations over $T = 250$ periods (result are similar with $T = 1000$). The VAR has 5 lags. For the test for non-fundamentalness, we project $\tilde{\varepsilon}$ on a constant and one lag or on four lags of ε and η . For each of those experiments, we run 10,000 simulations.

In the following Figures, we display the 95% bands and the median for the estimated responses of TFP and SP to an identified news shock, together with the theoretical response. The visual inspection of these IRF will indicate how severe non-fundamentalness is. We also present more formal measures. First, we plot the density of the p-value of the non-fundamentalness test. This test is the F-test that the coefficients of ε and η lags are all null.

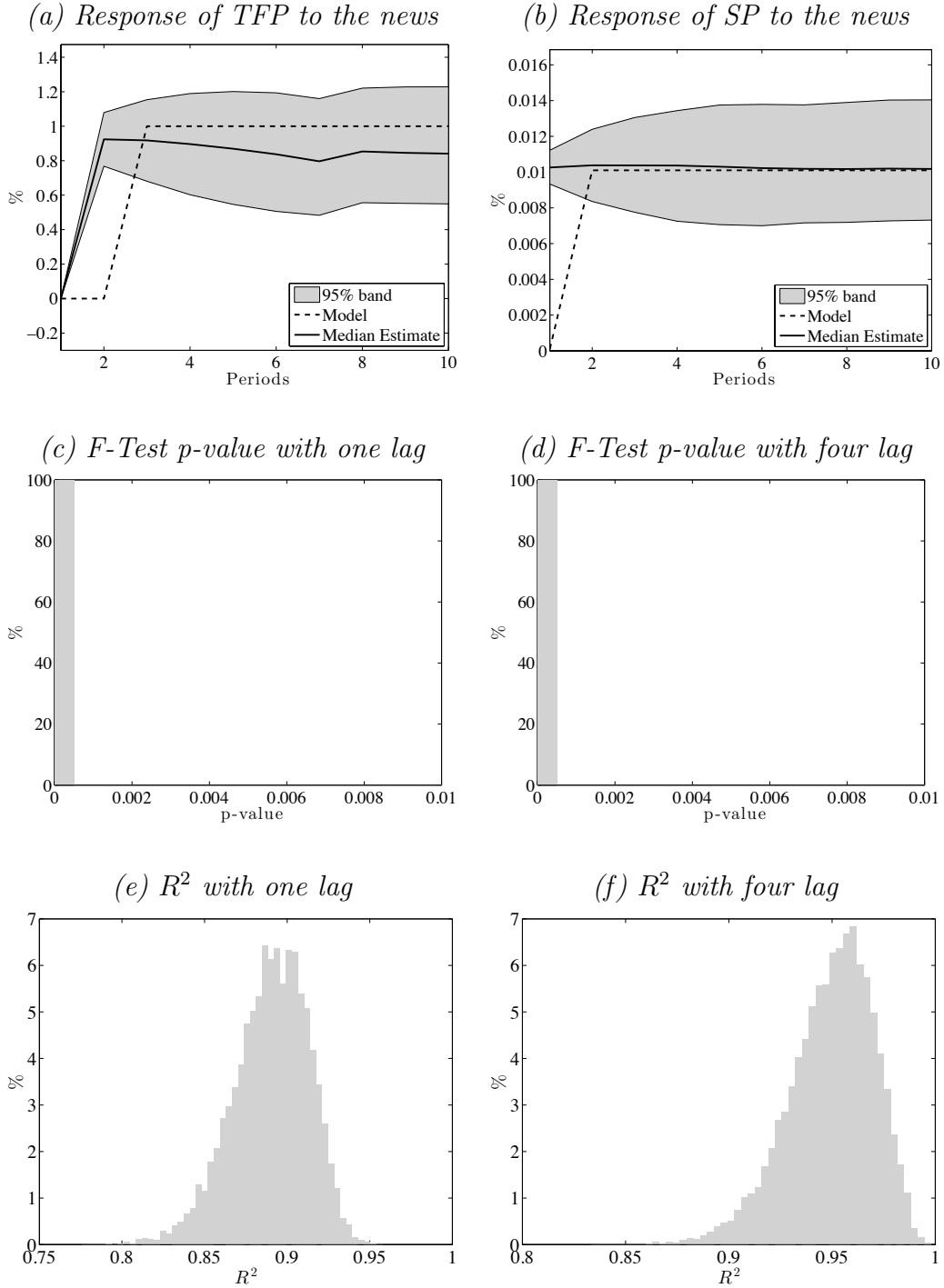
Second, we plot the density of the regression R^2 over the 10,000 simulations for each model. The R^2 is the one of the regression of $\tilde{\varepsilon}$ on a constant the past of ε and η . Note that the p-value will answer the question “Is there non-fundamentalness?” while the size of the R^2 will be more indicative of another question, namely “Does it matter?”. Results are presented in Figures 1 to 3 and are very telling.

When $\beta = .1$ (Figure 1), non-fundamentalness is a real problem. Estimated IRF are far from the theoretical ones (what matters are the two first periods, as the model responses are flat after and therefore quite easy to catch with the VAR). The p-value of the non-fundamentalness test are always equal to zero: fundamentalness is always rejected. Finally, R^2 s are close to 1: estimated shocks can be very well predicted by current theoretical innovations: non-fundamentalness does matter.

With a more realistic value of $\beta = .99$ typically used in quarterly models, SP data put much more weight on expectations of Θ in the data generating model. As shown in Figure 2, the theoretical non-fundamentalness now has little quantitative bite: estimated IRF are very close to that implied by the model. Nevertheless, the theoretical news shock is found non-fundamental when performing the F-test with four lags (although not with one lag). But even if fundamentalness is rejected, R^2 s are less than 0.1, which suggests that non-fundamentalness may not matter quantitatively, and which is confirmed by the inspection of the IRF.

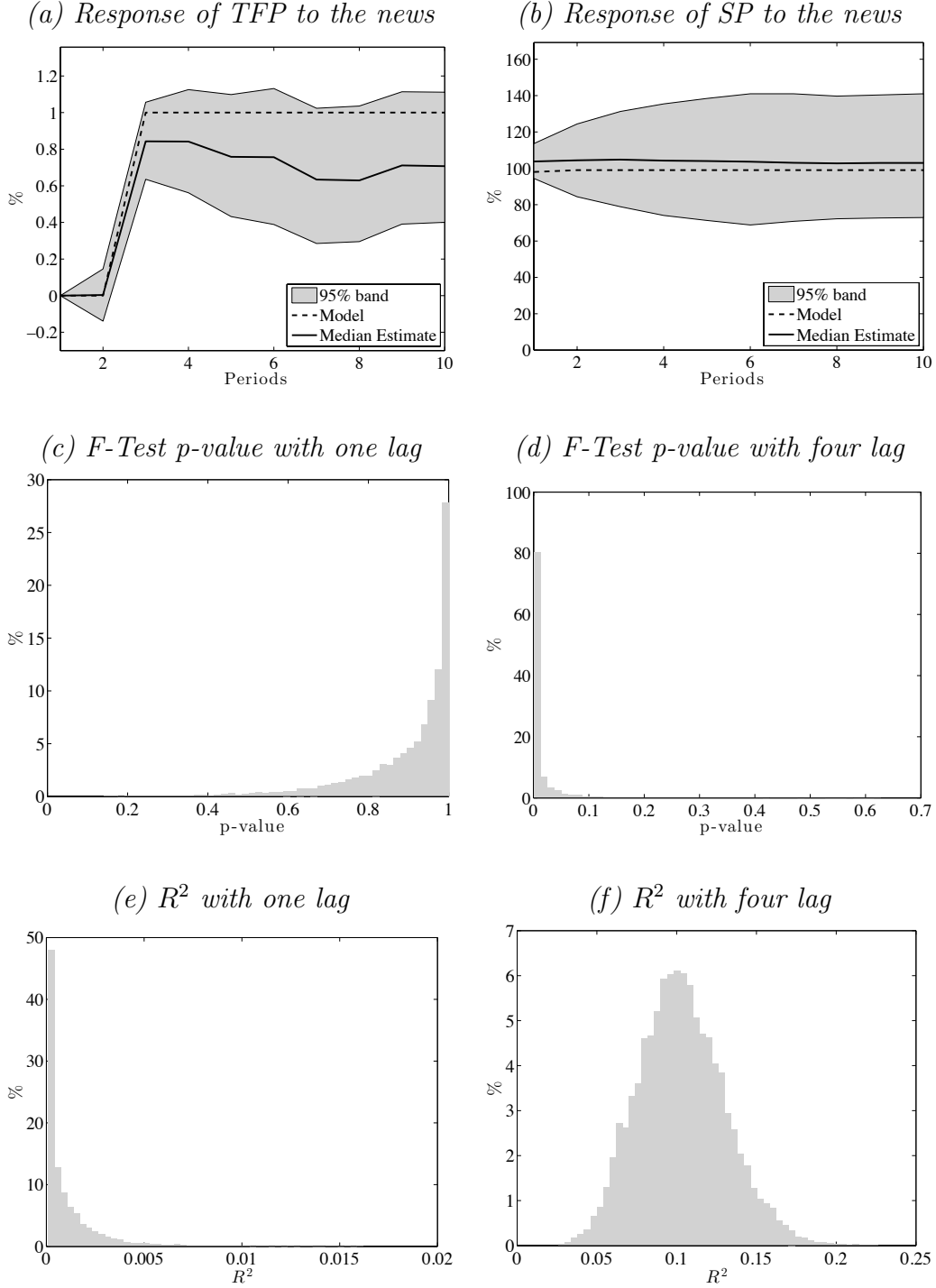
Finally, Figure 3 shows a case where R^2 are around .3. This is obtained for $\beta = .7$. The IRF are very well captured, except for a blip in TFP in period 2. Nevertheless, testing for fundamentalness will systematically reject it, as the p-values of the F-test are always less than 1%, and are actually zero more than 95% of the simulations. How can we explain that IRF are so well estimated and nevertheless fundamentalness rejected? This arises because the R^2 s of the regression of $\tilde{\varepsilon}$ on past values of the ε s are small (around .1 with one lag, around .3 with four lags).

Figure 1: A case with serious non-fundamentallness ($\beta = .1$)



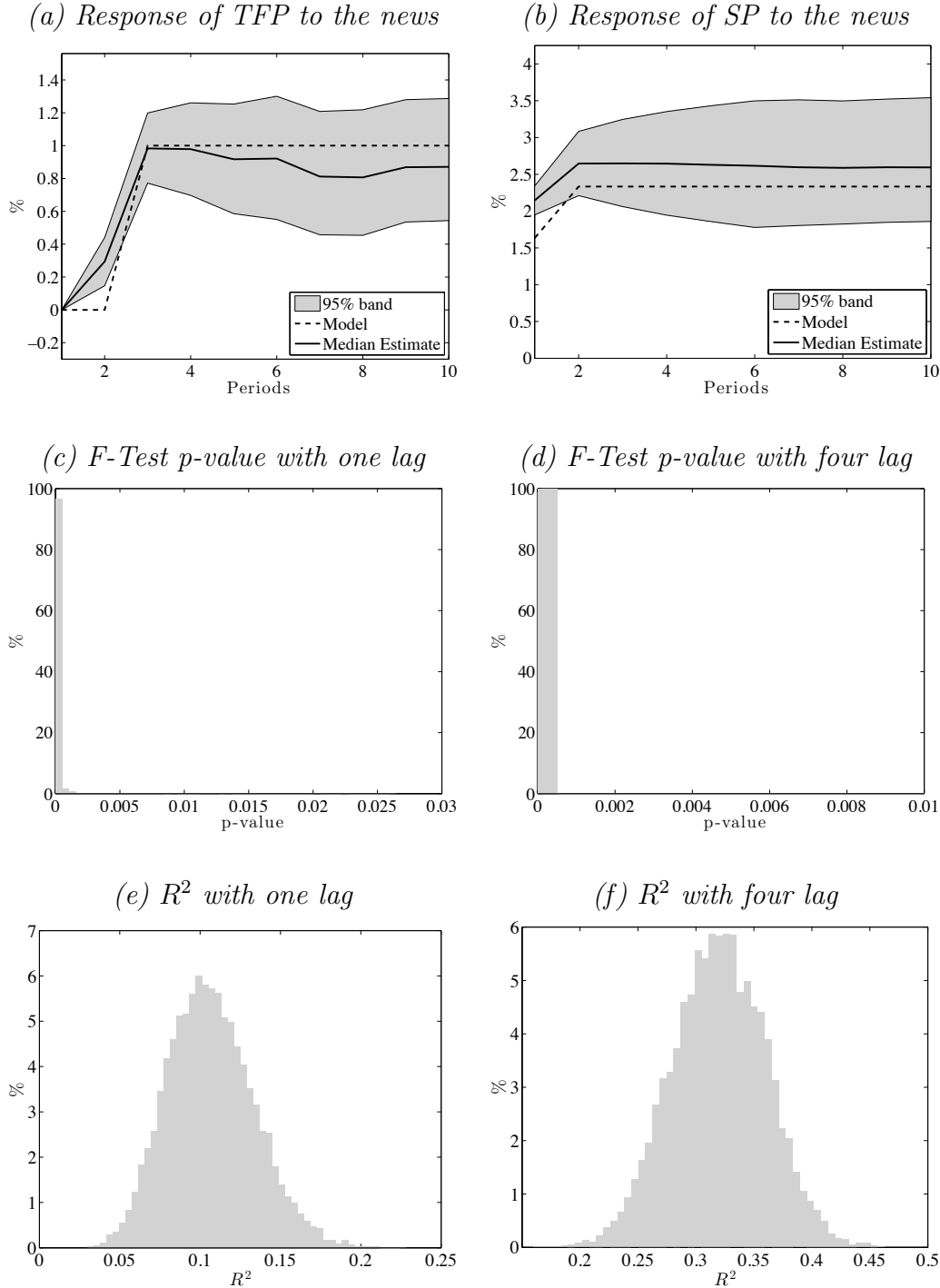
This figures are derived from the simulation of a simple Lucas' tree model with news. In panel (a) and (b) are represented the theoretical responses to a news shock (dashed lines) together with the median and 95% confidence interval obtained from the repeated estimation (10,000 replications) of a VAR 2 impulse response to the shock that does not affect TFP on impact. The estimated news shock is then projected on one or four lags of the two theoretical shocks, and we test for orthogonality. Panels (c) and (d) show the distribution of the p-value for the orthogonality test. Panels (e) and (f) show the distribution of the R^2 s of those regressions.

Figure 2: A case with no serious non-fundamentallness ($\beta = .99$)



This figures are derived from the simulation of a simple Lucas' tree model with news. In panel (a) and (b) are represented the theoretical responses to a news shock (dashed lines) together with the median and 95% confidence interval obtained from the repeated estimation (10,000 replications) of a VAR 2 impulse response to the shock that does not affect TFP on impact. The estimated news shock is then projected on one or four lags of the two theoretical shocks, and we test for orthogonality. Panels (c) and (d) show the distribution of the p-value for the orthogonality test. Panels (e) and (f) show the distribution of the R^2 s of those regressions.

Figure 3: A case with R^2 around .1 and .3 : non-fundamentalness is not much of problem ($\beta = .7$)



This figures are derived from the simulation of a simple Lucas' tree model with news. In panel (a) and (b) are represented the theoretical responses to a news shock (dashed lines) together with the median and 95% confidence interval obtained from the repeated estimation (10,000 replications) of a VAR 2 impulse response to the shock that does not affect TFP on impact. The estimated news shock is then projected on one or four lags of the two theoretical shocks, and we test for orthogonality. Panels (c) and (d) show the distribution of the p-value for the orthogonality test. Panels (e) and (f) show the distribution of the R^2 s of those regressions.

3 Application to the identification of TFP news shocks in U.S. data

In this section, we show that the results of Beaudry and Portier [2006] and [2014] are robust to the non-fundamentalness critique.

3.1 Baseline results

In the following, we use the same sample as used by Forni, Gambetti, and Sala [2014] and use the data described in Beaudry and Portier [2014].⁵ TFP is corrected for utilisation, consumption is total consumption (including durable) and investment is total investment (see data appendix for details).

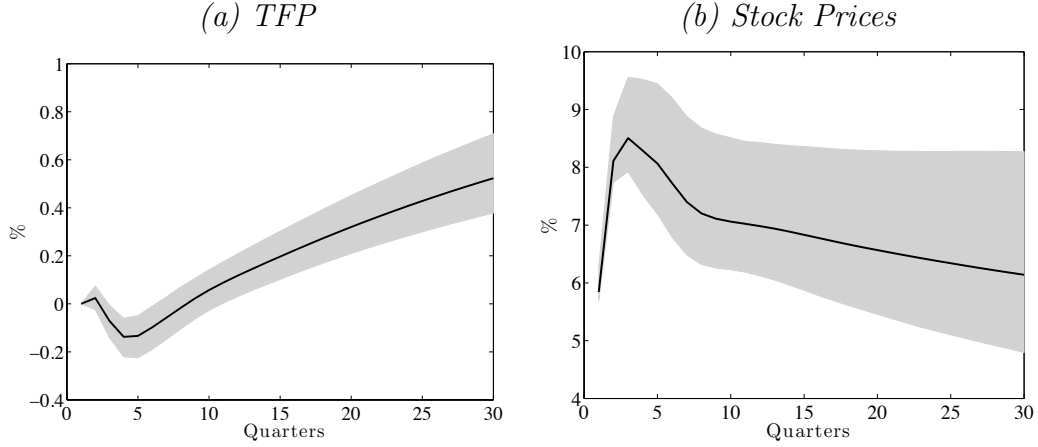
We first consider the basic Beaudry and Portier’s [2006] VAR 2. Whereas the small dimension of the VAR might be a weakness, this VAR has the advantage of being simple and, as we discussed in Beaudry and Portier [2014], gives results that are robust to various extensions. The two variables in the system are TFP and Stock Prices. The single identifying restriction is that the identified news has no impact effect on TFP, which correspond to a Choleski decomposition in which TFP is the first variable and the news shock the first shock. Figure 4 shows that we indeed identify a diffusion news. TFP does not increase for about 10 quarters⁶, but does in the long run.

We now extend the VAR to add three extra variables: consumption, investment and hours. To identify a TFP news shock, we follow the identification strategy set out in Beaudry and Portier [2014] which is a natural extension to that introduced in Beaudry and Portier [2006]. This identification strategy only identifies a news shock and an unrestricted technology shock, while the other shocks remain unnamed. The identifying restrictions are the following: (i) all the shocks but the unrestricted technology shock have zero impact effect on TFP, (ii) the news and the unrestricted technology shock are the only permanent shocks to TFP. In Beaudry and Portier [2014] we have shown that this identification gives robust results when one varies either the information set, the sample period and the specification.

⁵Note that the results of our VARs are robust to a longer sample (1946-2013), but the factors are only available on the shorter sample.

⁶TFP actually decreases, which might be the consequence of an excessive correction for utilization.

Figure 4: Response to a news shock in the Beaudry and Portier’s [2006] VAR 2



Data are described in the appendix and the sample period in 1960Q1-2012Q2. The news shock is the one that does not affect TFP on impact. The VAR is estimated in levels and with 4 lags. The unit of the vertical axis is percentage deviation from the situation without shock. Grey areas correspond to the 66% confidence band. The distribution of IRF is the Bayesian simulated distribution obtained by Monte-Carlo integration with 10,000 replications, using the approach for just-identified systems discussed in Doan [1992].

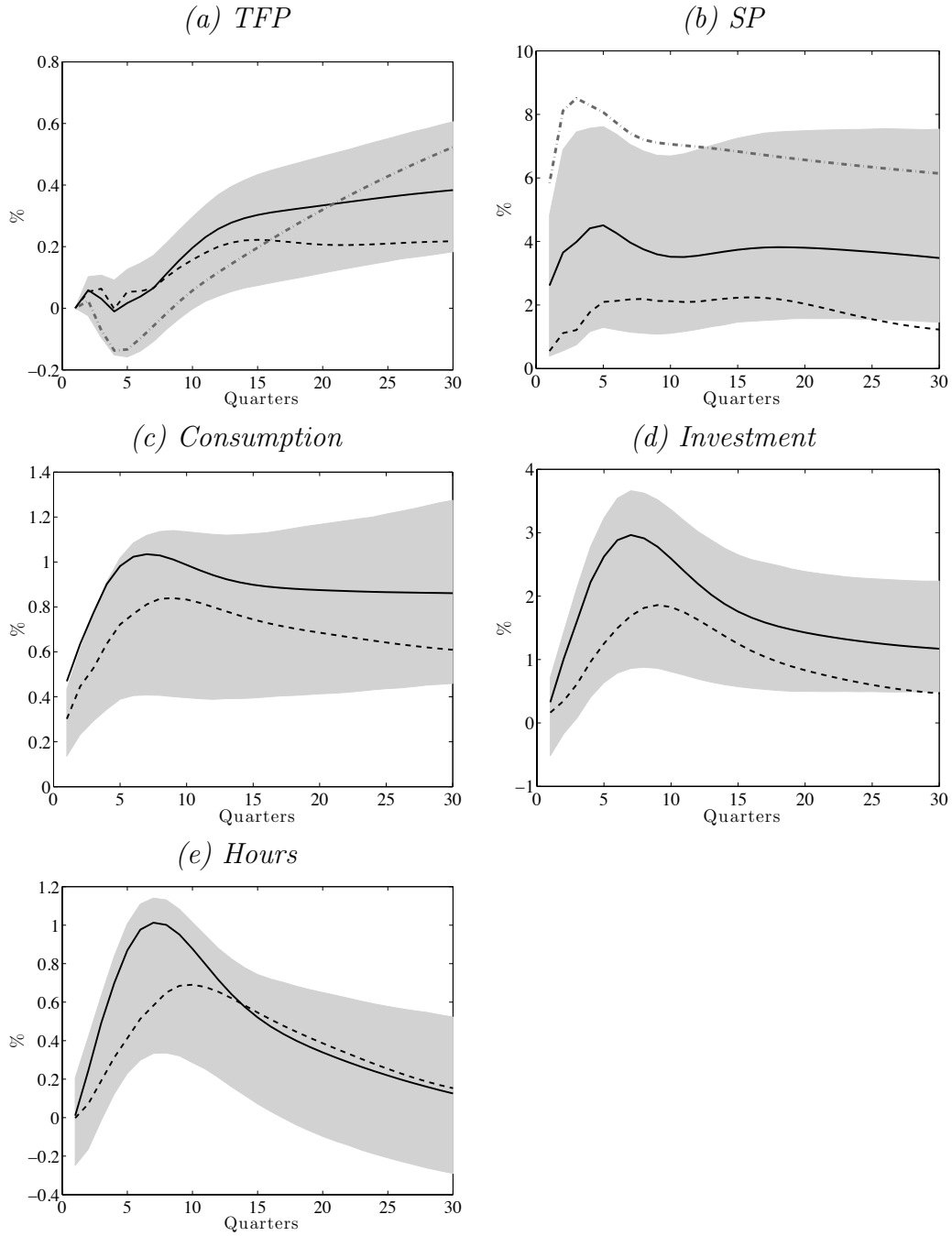
Impulse responses are presented in Figures 5. The plain line shows the point estimates. We observe all the characteristic of a news driven economic expansion. TFP does not move in the short run, the stock market reacts instantaneously to the news, consumption, investment and hours do increase on impact and subsequently, before any sizable increase in TFP. In panels (a) and (b), we also represent the responses of TFP and SP obtained from the VAR 2 (dashed-dotted gray line). Note that the response of TFP is very similar, while the responses of SP is now purged from some non-news related variations.

These results suggest that there are indeed news in the business cycle, but it might be the case, as pointed out by Forni, Gambetti, and Sala [2014], that the estimation suffers from non-fundamentalness. This is what we check now.

3.2 The quantitative unimportance of non-fundamentalness

In order to test for non-fundamentalness, we follow Forni, Gambetti, and Sala [2014]. The authors use a dataset composed of 107 US quarterly macroeconomic series, and estimate the principal components of this data set. They show that essentially all the information is contained in the first three factors. We therefore use these first three factors, which

Figure 5: Comparison of the VAR 5 responses with the ones of the VAR 5 augmented with the first three factors



Data are described in the appendix and the sample period is 1960Q1-2012Q2. In the VAR 5 (the plain line), the news shock is restricted to have no impact effect on TFP but is not restricted in the long run. The dotted lines correspond to the VAR 8, that is the VAR 5 augmented with the first three factors of Forni, Gambetti, and Sala [2014]. The dashed-dotted gray lines of panels (a) and (b) are the responses to a news shock in the VAR 2 of Figure 4. The VARs are estimated in levels and with 4 lags. The unit of the vertical axis is percentage deviation from the situation without shock. Grey areas correspond to the 66% confidence band of the VAR 5. The distribution of IRF is the Bayesian simulated distribution obtained by Monte-Carlo integration with 10,000 replications, using the approach for just-identified systems discussed in Doan [1992].

Forni, Gambetti and Sala kindly provided us. We project the estimated news shock of the VAR 2 and of the VAR 5 on one lag or four lags of the first three factors, and test for the orthogonality of our news shocks to the factors. The test is a F-test, and the p-values are reported in Table 1. In all cases, the p-value is less than 5%. We therefore do agree with Forni, Gambetti, and Sala [2014] that our identified strategy is likely subject to the non-fundamentality problem. However, does it matter for the estimation of the impulse response functions to a news shock? The answer we find is no, or at least not very much. A first element suggestive of this negative answer comes from the inspection of the R^2 s associated with specification test. These are displayed in Table 1. The R^2 s are never larger than .2: even though our estimated news shocks are not orthogonal to the factors, those factors explain less than 20% of the variance of the news. The simulation results of the previous section suggest that in such a case, the non-fundamentality should not be much of a quantitative problem.

We then re-estimate our VAR 5 by adding the three factors, so that we end up estimating a VAR 8. We use the same identification strategy, that is, : (i) all the shocks but the unrestricted technology shock have zero impact effect on TFP, (ii) the news and the unrestricted technology shock are the only permanent shocks to TFP. The estimated response to the newly identified news shock are the black dashed lines of Figure 5. Except for the Stock Price whose response has a similar shape but is divided by a factor two, the responses of TFP, consumption, investment and hours are all very similar to that obtained in the absence of including the factors. This contrasts with Forni, Gambetti, and Sala [2014] finding that is based on the identification strategy of Barsky and Sims [2011], which itself is not very supportive of the news shocks view of business cycles. Hence, these results suggest that our preferred means of identifying news shocks generate impulse responses with properties that are robust to the non-fundamentality critique.⁷ We therefore infer that the data still support the view that there likely is news in business cycles.

⁷ There may remain debate about how best to identify new shocks, but that is an issue entirely different from the issue of non-fundamentality emphasized in Forni, Gambetti, and Sala [2014]. See Beaudry, Nam, and Wang [2011] for some answers to that question.

Table 1: Test for non-fundamentalness and associated R^2 s

Model	One lag		Four lags	
	R^2	F-test p-value	R^2	F-test p-value
VAR 2	.03	.04	.18	.05
VAR 5	.09	.01	.21	.01

This Table presents the results of the non-fundamentalness test proposed by Forni, Gambetti, and Sala [2014]. For each VAR, the news shock is projected on one or four lags of the first three factors of Forni, Gambetti, and Sala [2014]. Table includes the p-value for the orthogonality test, as well as the R^2 of those regressions. Data are described in the appendix and the sample period in 1960Q1-2012Q2. In the VAR 2, the news shock is the one that does not affect TFP on impact. In the VAR 5, the news shock is only restricted to have no impact effect on TFP but is not restricted in the long run. The VARs are estimated in levels and with 4 lags.

4 Conclusion

Taking seriously the observation of Forni, Gambetti, and Sala [2014] that news shocks can be non-fundamental with respect to the information set of the econometrician, we have first shown using a simple Lucas' tree model with news that one should separate the very existence of non-fundamentalness from its quantitative importance. We have then shown that when estimating technological news shock with an identification scheme we have shown previously to be very robust, we found that non-fundamentalness is present, but that it does not matter quantitatively. This is not of course a general result that would apply to any VAR estimation. In fact, the test proposed by Forni, Gambetti, and Sala [2014] is a useful one that macro-econometricians should systematically perform when non-fundamentalness may be present. However, this tests should not only be used to detect non-fundamentalness, but the R^2 associated with the test should also be used to help assess whether non-fundamentalness may be quantitative important. For technological news shocks, our findings suggest that non-fundamentalness is not likely to be a quantitatively important issue.

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Data Appendix

- Hours: BLS, Series Id: PRS85006033, Nonfarm Business sector, 1947Q1-2012Q3, seasonally adjusted, downloaded: 12/2012
- Consumption: BEA, Table 1.1.3. Real Gross Domestic Product, Quantity Indexes, 1947Q1-2012Q3, seasonally adjusted, downloaded: 12/2012
- Investment: BEA, Table 1.1.3. Real Gross Domestic Product, Quantity Indexes, 1947Q1-2012Q3, seasonally adjusted, downloaded: 12/2012
- TFP: Utilization-adjusted quarterly-TFP series for the U.S. Business Sector, produced by John Fernald, series ID: dtfp_util, 1947Q1-2012Q3, downloaded: 12/2012
- Stock Prices: S&P500 index deflated by CPI, obtained from the [homepage of Robert J. Shiller](#).

On The Quantitative Importance of Non-fundamentalness for News Shocks

Extra Material

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October 2014

This document gathers some extra material related to the paper “On The Quantitative Importance of Non-fundamentalness for News Shocks”.

1 Comparison between Beaudry and Portier’s [2014] and Forni, Gambetti, and Sala’s [2014] identification

Here we compare the IRF of two different VARs and two different identifications:

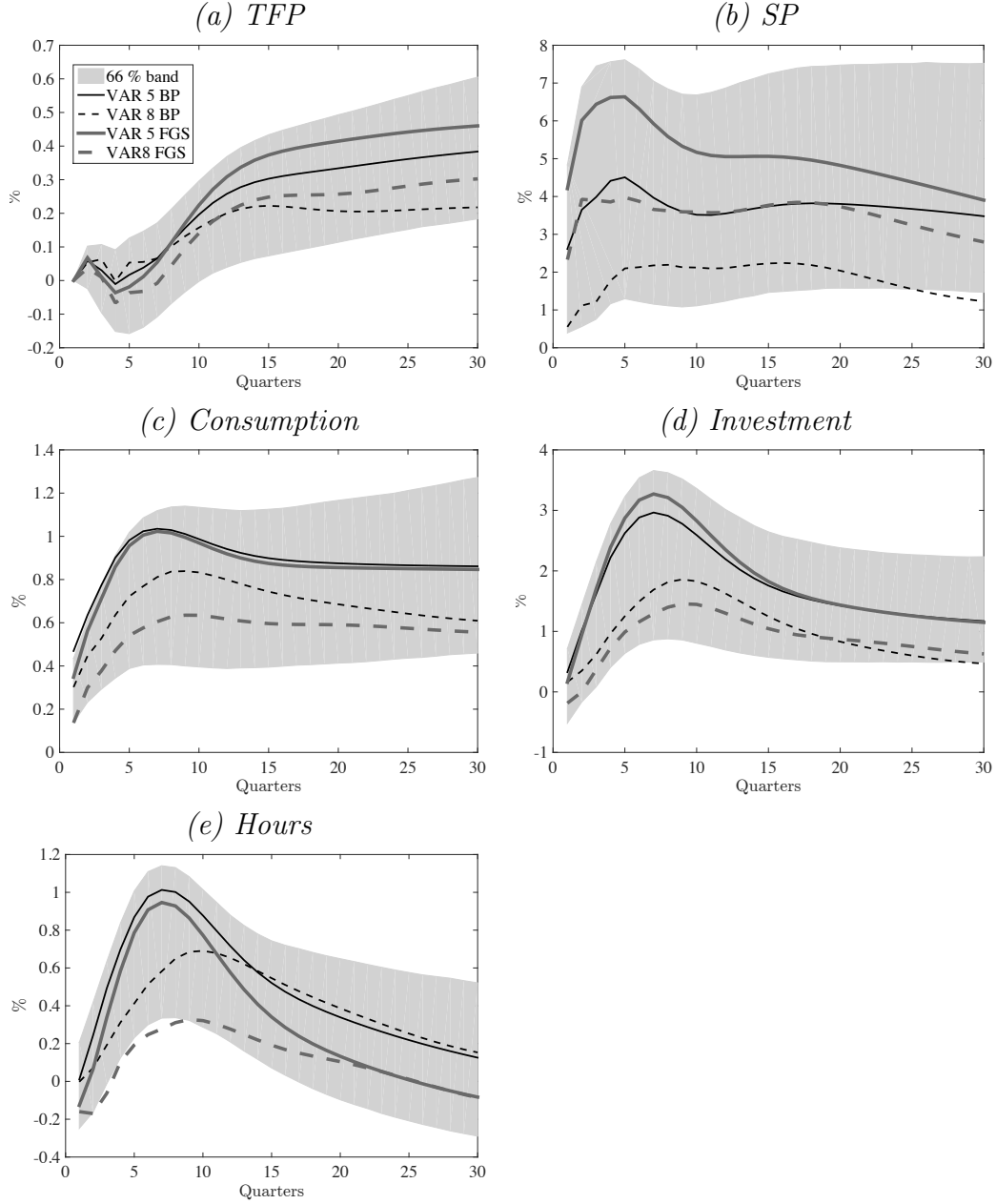
1. A VAR 5 with TFP, SP, C, I and H with Forni, Gambetti, and Sala’s [2014] identification.
2. A VAR 5 with TFP, SP, C, I and H with Beaudry and Portier’s [2014] identification.
3. A VAR 8 with TFP, SP, C, I and H plus the three first factors of Forni, Gambetti, and Sala [2014] with Forni, Gambetti, and Sala’s [2014] identification.
4. A VAR 8 with TFP, SP, C, I and H plus the three first factors of Forni, Gambetti, and Sala [2014] with Beaudry and Portier’s [2014] identification.

Impulse responses and variance shares are not significantly different from the VAR 5 with Beaudry and Portier’s [2014] identification when using the three factors or with the Forni, Gambetti, and Sala’s [2014] identification.

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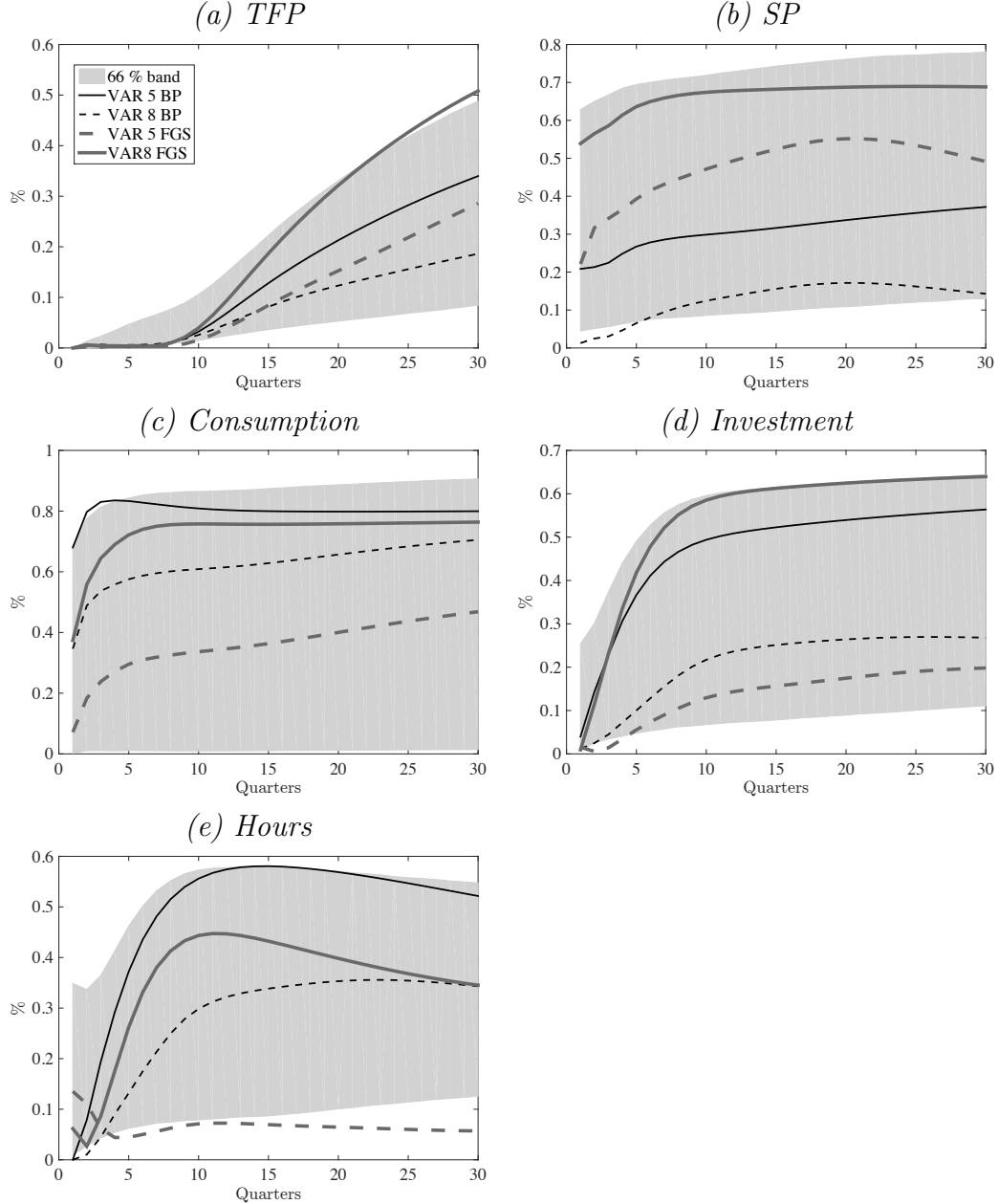
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Figure 1: Response to a News Shock, Comparison between Beaudry and Portier's [2014] and Forni, Gambetti, and Sala's [2014] identification



Data are described in the appendix and the sample period is 1960Q1-2012Q2. The plain lines correspond to the VAR 5, the dashed lines to the VAR 8, that is the VAR 5 augmented with the first three factors of Forni, Gambetti, and Sala's [2014]. The black lines correspond to Beaudry and Portier's [2014] identification and the gray lines to Forni, Gambetti, and Sala's [2014] one. The VARs are estimated in levels and with 4 lags. The unit of the vertical axis is percentage deviation from the situation without shock. Grey areas correspond to the 66% (95% for consumption) confidence band of the VAR 5 with Beaudry and Portier's [2014] identification. The distribution of IRF is the Bayesian simulated distribution obtained by Monte-Carlo integration with 10,000 replications, using the approach for just-identified systems discussed in Doan [1992].

Figure 2: Share of the Forecast Error Variance Attributable to the News Shock, Comparison between Beaudry and Portier's [2014] and Forni, Gambetti, and Sala's [2014] identification



Data are described in the appendix and the sample period is 1960Q1-2012Q2. The plain lines correspond to the VAR 5, the dashed lines to the VAR 8, that is the VAR 5 augmented with the first three factors of Forni, Gambetti, and Sala's [2014]. The black lines correspond to Beaudry and Portier's [2014] identification and the gray lines to Forni, Gambetti, and Sala's [2014] one. The VARs are estimated in levels and with 4 lags. The unit of the vertical axis is percentage deviation from the situation without shock. Grey areas correspond to the 66% confidence band of the VAR 5 with Beaudry and Portier's [2014] identification. The distribution of IRF is the Bayesian simulated distribution obtained by Monte-Carlo integration with 10,000 replications, using the approach for just-identified systems discussed in Doan [1992].

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